

MATH2099 Notes

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Python

Preamble

```
import numpy as np
import scipy as sci
import pandas as pd
import matplotlib.pyplot as plt
```

Linear Algebra

Definitions

Vector definition	$\begin{pmatrix} x \\ y \\ z \\ \dots \end{pmatrix}$	<code>np.array([x, y, z, ...])</code>
Matrix definition	$\begin{pmatrix} v_1 & v_2 & \dots & v_n \\ v_4 & v_5 & \dots & v_6 \\ \vdots & \vdots & \ddots & \vdots \\ v_7 & v_8 & \dots & v_9 \end{pmatrix}$	<code>np.array([[v1, v2, ..., v3], [v4, v5, ..., v6], [..., ..., ..., ...], [v7, v8, ..., v9]])</code>
Matrix augmentation	$(v_1 \mid v_2 \mid \dots \mid v_n)$	<code>np.column_stack((v1, v2, ..., v3))</code>
	$\begin{pmatrix} v_1 \\ v_2 \\ \dots \\ v_n \end{pmatrix}$	<code>np.row_stack((v1, v2, ..., v3))</code>
Zero matrix / vector	$\begin{pmatrix} 0 \\ 0 \\ \dots \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix}$	<code>np.zeros(n)</code> <code>np.zeros((m, n))</code>
Identity matrix	$I \in \mathbb{C}^n$	<code>np.eye(n)</code>
Diagonal matrix	$\begin{pmatrix} a & 0 & \dots & 0 \\ 0 & b & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & c \end{pmatrix}$	<code>np.diag([a, b, ..., c])</code>

Vector element access	v_i	<code>v[i]</code> # 0-indexed
Matrix row access	M_i	<code>M[i]</code> # 0-indexed
Matrix column access	$M_{:,i}$	<code>M[:, i]</code> # 0-indexed
Matrix element access	M_{ij}	<code>M[i, j]</code> # 0-indexed

Vector dot product	$a \cdot b$	<code>np.dot(a, b)</code> or <code>a.dot(b)</code>
Matrix multiplication	AB	<code>A @ B</code>
Matrix determinate	$\det(A)$	<code>np.linalg.det(A)</code>
Matrix inverse	$\text{inv}(A)$	<code>np.linalg.inv(A)</code>
Matrix transpose	A^T	<code>A.T</code>
Solve linear equation	$Ax = b$	<code>x = np.solve(A, b)</code>

Eigen value / vector	$\text{eig}(\mathbf{A})$	<code>np.linalg.eig(A)</code> <code>np.linalg.eigvals(A)</code>
Matrix rank	$\text{rank}(\mathbf{A})$	<code>np.linalg.matrix_rank(A)</code>
Matrix nullspace	$\text{nullspace}(\mathbf{A})$	<code>sci.linalg.null_space(A)</code>
Matrix norm	$\text{norm}(\mathbf{A})$	<code>np.linalg.norm(A, ord=?)</code>

Statistics

Import Data	<code>pd.read_csv("filename.txt", delimiter="\t")</code>
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Mean	<code>np.mean(a)</code> <code>dataframe.mean()</code>
Median	<code>np.median(a)</code> <code>dataframe.median()</code>
Standard deviation	<code>np.std(a, ddof=1)</code> <code>dataframe.std()</code>
Variance	<code>np.var(a, ddof=1)</code> <code>dataframe.var()</code>
Quartiles	<code>np.quantiles(a, [0, 0.25, 0.5, 0.75, 1], method='hazen')</code> # For min, q1, median, q3, max
Skewness	<code>sc.stats.skew(a)</code>

Plotting

Scatter Plot	<code>dataframe.plot.scatter(xcol, ycol)</code>
Line Plot	<code>dataframe.plot.line()</code>
Histogram	<code>dataframe.plot.hist(bins) # bins: int []int</code>
Density Histogram	<code>dataframe.plot.hist(density=True, bins)</code>

Binomial

$X \sim \text{Bin}(n, p)$	
$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$	<code>sc.stats.binom(n, p)</code>
$E[X] = np, \text{Var}(X) = np(1 - p)$	

Poisson

$X \sim \text{Poi}(\lambda)$	
$P(X = k) = \lambda^k \frac{e^{-\lambda}}{k!}$	<code>sc.stats.poisson(lam)</code>
$E[X] = \lambda, \text{Var}(X) = \lambda$	

Exponential

$X \sim \text{Exp}(\lambda)$	
$f(x) = \lambda e^{-\lambda x}$	<code>sc.stats.expon(scale=1/lam)</code>
$E[X] = \frac{1}{\lambda}, \text{Var}(X) = \frac{1}{\lambda^2}$	

Normal

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$$

$$E[X] = \mu, \text{Var}(X) = \sigma^2$$

sc.stats.norm(mu, sigma)

Uniform

$$X \sim \text{Uniform}(a, b)$$

$$f(x) = \frac{1}{b-a} \text{ for } x \in [a, b]$$

$$E[X] = \frac{a+b}{2}, \text{Var}(X) = \frac{(b-a)^2}{12}$$

sc.stats.uniform(loc=a, scale=b-a)

Given dist = sc.stats.xxx(...)

.pmf(k) = $P(X = k)$ (discrete)

.pdf(x) = $f(x)$ (continuous)

.cdf(x) = $P(X \leq x)$

.ppf(q) = $F^{-1}(q)$ (quantile)

.mean(), .var(), .std()

Linear Algebra

Linear Equations and Matrices

Vector dot product

$$\vec{a} \cdot \vec{b} = \sum_{i=1}^n a_i b_i$$

Matrix Vector Multiplication

$$A\vec{v} = (a_1 \mid a_2 \mid \dots \mid a_n)\vec{v} = a_1 v_1 + a_2 v_2 + \dots + a_n v_n$$

Matrix Matrix Multiplication

$$AB = (A\vec{b}_1 \mid A\vec{b}_2 \mid \dots \mid A\vec{b}_n)$$

Matrices Properties

- $A + B = B + A$
- $(A + B) + C = A + (B + C)$
- $A + 0 = A$
- $A + (-A) = 0$
- $\lambda(\mu A) = (\lambda\mu)A$
- $(\lambda + \mu)A = \lambda A + \mu A$
- $\lambda(A + B) = \lambda A + \lambda B$
- $A(BC) = (AB)C$
- $A(B + C) = AB + AC$
- $A(\lambda B) = \lambda(AB) = A(\lambda B)$
- $A0 = 0A = 0$
- $AI + IA = A$
- $AB \neq BA$ (usually)

Transpose A^T = diagonally flipped from top left to bottom right

Inverse Matrix X is the inverse of A if $AX = XA = I$

Transpose and Inverse Properties:

- $(A^T)^T = A$
- $(\lambda A + \mu B)^T = \lambda A^T + \mu B^T$
- if AB exists, then $(AB)^T = B^T A^T$
- inverses only exist for square matrices $M \in C_{n,m}$ for $n = m$
- If A is invertible the $(A^{-1})^{-1} = A$
- If both A and B are invertible and are the same shape, $(AB)^{-1} = B^{-1} A^{-1}$

Symmetric $A^T = A$

Skew-symmetric $A^T = -A$

Orthogonal $A^T A = A A^T \Leftrightarrow A^T = A^{-1}$

Vector Spaces

Vector Space A vector space V over the field of scalars $\mathbb{F}$ is a non-empty set of **vectors** for which addition of vectors and multiplication of a vector by a scalar are defined and obey the following axioms.

1. Closure under addition. If $u, v \in V$ then $u + v \in V$.
2. Associative law of addition. If $u, v, w \in V$ then $(u + v) + w = u + (v + w)$.
3. Commutative law of addition. If $u, v \in V$ then $u + v = v + u$.
4. Existence of zero. There is a special element 0 in V called the **zero vector** which has the property that $v + 0 = v$ for all $v \in V$.
5. Existence of Negative: For each $v \in V$ there exists an element $w \in V$ such that $v + w = 0$.
6. Closure under scalar multiplication. If $v \in V$ and $\lambda \in \mathbb{F}$, then $\lambda v \in V$.
7. Associative law of multiplication by a scalar. If $\lambda, \mu \in \mathbb{F}$ and $v \in V$ then $\lambda(\mu v) = (\lambda\mu)v$
8. Behaviour of scalar 1. If $v \in V$ then $1v = v$.
9. Scalar distributive law. If $\lambda, \mu \in \mathbb{F}$ and $v \in V$ then $(\lambda + \mu)v = \lambda v + \mu v$.
10. Vector distributive law. If $\lambda \in \mathbb{F}$ and $u, v \in V$ then $\lambda(u + v) = \lambda u + \lambda v$.

Important Vector Spaces

- The vector spaces $\mathbb{R}^n$ over $\mathbb{R}$, and $\mathbb{C}^n$ over $\mathbb{C}$.
- The vector space of all polynomials of degree n or less with coefficients in $\mathbb{F}$, $\mathbb{P}_n(\mathbb{F})$ over $\mathbb{F}$.
- The set of all polynomials with coefficients in $\mathbb{F}$, $\mathbb{P}(\mathbb{F})$ over $\mathbb{F}$.
- The vector space of all $p \times q$ matrices, $M_{p,q}(\mathbb{F})$ over $\mathbb{F}$.
- The set of all real-valued functions with domain X , $R[X]$ over $\mathbb{R}$.

Subspaces

Subspace A subset S of a vector space V is called a **subset** of V if S is itself a vector space over the same field of scalars as V and under the same rules for addition and multiplication by scalars.

In addition, if S is a proper subset of V (i.e. $S \neq V$), S is called a **proper subspace** of V .

The set $\{0\}$ consisting of just the zero vector in V is always a subspace of V and is called the **trivial subspace** of V .

Proving Subspace (Subspace Theorem)

A subset S of a vector space V is a subspace if and only if:

- the zero vector of V belongs to S

- $u + v \in S$ for all $u, v \in S$
- $\lambda v \in S$ for all $v \in S$, scalars λ

Linear Combinations and Spans

Linear Combination Definition Let $S = \{v_1, \dots, v_n\}$ be a subset of a vector space V over a field $\mathbb{F}$. A **linear combination** of S is a sum of scalar multiples of the form

$$\lambda_1 v_1 + \dots + \lambda_n v_n \text{ with } \lambda_1, \dots, \lambda_n \in \mathbb{F}$$

Span Let $S = \{v_1, \dots, v_n\}$ be a subset of a vector space V . Then the **span** of the set S is the set of all linear combinations of S , and is denoted by $\text{span}(S)$ or $\text{span}(v_1, \dots, v_n)$.

If $\text{span}(S) = V$, the set S is called the **spanning set** of V , and S is said to **span** V or **generate** V .

Linear Independence

Linear Independence Given a set of vectors $S = \{v_1, \dots, v_n\}$. If the **only** solution of

$$\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n = 0$$

is the **trivial one** i.e. $\lambda_1 = \lambda_2 = \dots = \lambda_n = 0$, then we say that S is a **linearly independent set** and that the vectors in S are **linearly independent**.

If we can find scalars $\lambda_1, \lambda_2, \dots, \lambda_n$ **not all zero** such that

$$\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n = 0,$$

then we say that S is a **linearly dependent set** and that the vectors in S are **linearly dependent**.

Note: Let S be a finite set of vectors in $\mathbb{R}^p$ and A the matrix with the vectors in S as **columns** in order.

Then S is linearly independent if and only if $Ax = 0$ has the unique solution $x = 0$.

Basis and Dimension

Basis Definition A set B of vectors in a vector space V is called a **basis** for V if B is linearly independent and $V = \text{span}(B)$.

Theorem (Dimension Exists): If a vector space V has a finite spanning set then it has a finite basis and every basis of V has the same number of elements.

In this case, we call the number of vectors in a basis of V the **dimension** of V and write it as $\dim(V)$.

Standard basis: For $\mathbb{R}^n$ or $\mathbb{C}^n$ the **standard basis**, usually denoted $\mathcal{S}$ is the set of vectors $\{e_1, \dots, e_n\}$ where e_i has all entries zero except the i th entry which is 1. It follows that $\dim \mathbb{R}^n = n$.

For $\mathbb{P}_n$, the **standard basis**, also usually denoted $\mathcal{S}$, is the set of polynomials $\{1, t, t^2, \dots, t^n\}$.
 $\dim \mathbb{P}_n = n + 1$

Theorem: Let V be a vector space. If S is a finite spanning set of V and T is a linearly independent set in V , then

$$|T| \leq \dim(V) \leq |S|$$

Furthermore

- if $|T| = \dim(V)$, T is a basis for V
- if $|S| = \dim(V)$, S is a basis for V

Coordinate Vectors

Coordinate Vectors Suppose that $B = \{b_1, b_2, \dots, b_n\}$ is a basis for a vector space V where we fix the order of these vectors.

Let v be a vector in V . The fact that B spans V tell us that

$$v = x_1 b_1 + x_2 b_2 + \dots + x_n b_n$$

for certain scalars $x_1, x_2, \dots, x_n$; the fact that B is linearly independent tells us that for any specific v , there is only **one** possible choice of these scalars.

The scalars x_i are called the **coordinates** of the vector v with respect to the basis B .

Linear Maps

Linear Map Let V and W be two vector spaces over the same field $\mathbb{F}$. A function $T : V \rightarrow W$ is called a **linear map**, a **linear transformation** or described as **linear** if the following two conditions are satisfied:

- Addition is respected

$$T(u + v) = T(u) + T(v) \text{ for all } u, v \in V$$

- Scalar multiplication is respected

$$T(\lambda v) = \lambda T(v) \text{ for all } \lambda \in \mathbb{F} \wedge v \in V$$

Linearity Test Theorem A function $T : V \rightarrow W$ is a linear map if and only if for all $\lambda, \mu \in \mathbb{F}$ and $u, v \in V$

$$T(\lambda u + \mu v) = \lambda T(u) + \mu T(v)$$

If $T(0) \neq 0$, or there is a vector v with $T(-v) \neq -T(v)$ then T is not linear.

Matrices of Linear Transformations

Linear transformations are very close and have an important relationship to matrix multiplication. e.g.

$$\begin{aligned}
T \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} &= \begin{pmatrix} 7x_1 + x_2 \\ x_2 + 4x_3 \end{pmatrix} \\
&= \begin{pmatrix} 7 & 1 & 0 \\ 0 & 1 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}
\end{aligned}$$

We say that $A = \begin{pmatrix} 7 & 1 & 0 \\ 0 & 1 & 4 \end{pmatrix}$ is the matrix of T with respect to standard bases.

Essentially

$$T(x) = Ax.$$

Matrix of a Linear Map Let $T : V \rightarrow W$ be a linear transformation, where the domain V and the codomain W are finite-dimensional vector space over field $\mathbb{F}$.

Then A is the matrix of T with respect to ordered bases B for V and C for W if

$$[T(v)]_C = A[v]_B$$

for all vectors $v \in V$.

Here $[v]_B$ denotes the coordinate vector of v with respect to the basis B , and $[T(v)]_C$ denotes the coordinate vector of $T(v)$ with respect to the basis C .

A transformation matrix $A \in M_{q,p}$ is a linear map from a p dimensional vector space to a q dimensional has a $q \times p$ matrix.

Alternative method of finding the matrix of a linear transformation:

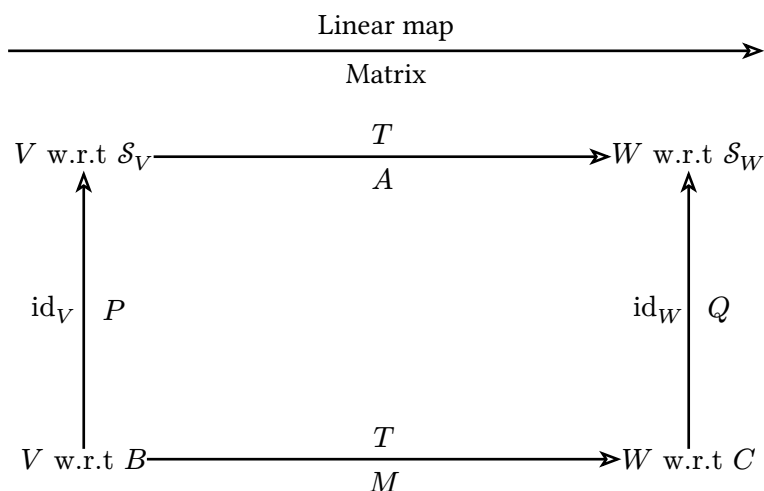
- Calculate the $T(v)$ for all v in the basis vector.
- Combine each transformed vector columnwise to find the transformation matrix.

Commutative Diagrams and Matrices of Linear Transformations

Lemma 1 If P is the matrix for $T_1 : U \rightarrow V$ with respect to bases B and C' and Q is the matrix for $T_2 : V \rightarrow W$ with respect to bases C' and C . Then QP is the matrix for $T_2 \circ T_1$ with respect to bases B and C .

Lemma 2 Let $T : V \rightarrow W$ be a linear transformation where $\dim(V) = \dim(W)$. Suppose M is the matrix of T with respect to bases B in V and C in W .

Then T is a bijection if and only if M is invertible, and in that case M^{-1} is the matrix of $T^{-1} : W \rightarrow V$ with respect to the bases C in W and B in V .



We call this a **commutative** diagram because you get the same result whichever path we take from one corner to another.

e.g.

$$\begin{aligned}\text{id}_W \circ T &= T \circ \text{id}_V \\ T &= Q^{-1} \circ T \circ P\end{aligned}$$

by the two lemmas, this means that

$$\begin{aligned}QM &= AP \\ M &= Q^{-1}AP\end{aligned}$$

w.r.t with respect to the basis

Kernal and Image

Kernal and Image of a Linear Transformation Let $T : V \rightarrow W$ be a linear map.

The **kernal** (or **nullspace**) of T , written $\ker(T)$, is the set of all zeros of T , that is,

$$\ker(T) = \{v \in V : T(v) = 0\}.$$

The **image** of T is the set of all function values of T , that is,

$$\text{im}(T) = \{w \in W : w = T(v) \text{ for some } v \in V\}.$$

Kernal and Image of a Matrix The **kernal** of a $p \times q$ matrix A is the subset of $\mathbb{R}^q$ defined by

$$\ker(A) = \{x \in \mathbb{R}^q : Ax = 0\}.$$

The **image** (or **nullspace**) of a $p \times q$ matrix A is the subset of $\mathbb{R}^p$ defined by

$$\text{im}(A) = \{b \in \mathbb{R}^p : b = Ax \text{ for some } x \in \mathbb{R}^q\}.$$

Rank and Nullity The **rank** of a linear map T is the dimension of $\text{im}(T)$ and the **rank** of a matrix A is the dimension of $\text{im}(A)$, denoted by $\text{rank}(T)$ and $\text{rank}(A)$.

The **nullity** of a linear map T is the dimension of $\ker(T)$ and the **nullity** of a matrix A is the dimension of $\ker(A)$, denoted by $\text{nullity}(T)$ and $\text{nullity}(A)$.

Rank-Nullity Theorem If A is an $p \times q$ matrix, then $\text{rank}(A) + \text{nullity}(A) = q$.

If $T : V \rightarrow W$ is a linear map between finite dimensional vector spaces

$$\text{rank}(T) + \text{nullity}(T) = \dim(V).$$

Some Linear Transformations on $\mathbb{R}^2$ and $\mathbb{R}^3$

Transformation

Matrix

$\mathbb{R}^2$

Rotation by θ

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

Reflection across x -axis

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Reflection across y -axis

$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

Reflection across $y = x$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Scaling by k

$$\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$$

Horizontal shear by k

$$\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$$

Vertical shear by k

$$\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}$$

Projection onto x -axis

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

Projection onto y -axis

$$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$\mathbb{R}^3$

Rotation about z -axis

$$\begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Rotation about x -axis

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}$$

Rotation about y -axis

$$\begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}$$

Reflection across xy -plane

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

Scaling by k

$$\begin{pmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & k \end{pmatrix}$$

Projection onto xy -plane

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Statistics

Revision

Definitions

Random experiment any experiment where the outcome cannot be predicted

Outcome the result of a random experiment

Event a set of possible outcomes

Sample space the set containing all possible outcomes

Probability of an event E $P(E)$ is the number measuring how likely event E is to occur from a random experiment.

Union $A \cup B$ the probability that either A or B occurs

Intersection $A \cap B$ the probability both A and B occur

Complement A^C the probability that A does not occur

$A \subseteq B$ A implies B

DeMorgan's $(A \cup B)^C = A^C \cap B^C$ and $(A^C \cap B^C) = (A \cup B)^C$

Probability Rules

- If $A \cap B = \emptyset$, A and B are mutually exclusive events.
- For any event E , $0 \leq \mathbb{P}(E) \leq 1$
- $\mathbb{P}(S) = 1$
- For any sequence of mutually exclusive events E_n , $\mathbb{P}\left(\bigcup_{i=0}^n E_i\right) = \sum_{i=0}^n \mathbb{P}(E_i)$
- $\mathbb{P}(E^C) = 1 - \mathbb{P}(E)$
- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$

Conditional Probability

- The conditional probability of B conditional on A is $\mathbb{P}(B | A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)}$
- If $B \subseteq A$, then $\mathbb{P}(A | B) = 1$
- $\mathbb{P}(A \cap B) = \mathbb{P}(A | B) \times \mathbb{P}(B)$
- Law of total probability: $\mathbb{P}(A) = \mathbb{P}(A | B) \times \mathbb{P}(B) + \mathbb{P}(A | B^C) \times \mathbb{P}(B^C)$
- Bayes' rules: If $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$ then $\mathbb{P}(A | B) = \mathbb{P}(B | A) \times \frac{\mathbb{P}(A)}{\mathbb{P}(B)}$

Event Independence

- Two events are independent iff $\mathbb{P}(A \cap B) = \mathbb{P}(A) \times \mathbb{P}(B)$
- Independence implies that $\mathbb{P}(A | B) = \mathbb{P}(A)$ and $\mathbb{P}(B | A) = \mathbb{P}(B)$

Introduction and descriptive statistics

Statistics is the science of the collection, processing, analysis, and interpretation of data

Population Obtain information about the entire collection of elements

Sample A subset of the population

Only sampling scheme that guarantees the sample to be a representative of the population is **random sampling**. Information drawn in a non-random sample cannot be generalised to larger populations.

Graphical Summary

Variable Types

One Categorical

One Quantitative

Graphical Plot

Histogram

Boxplot / Histogram

Variable Types

Two Categorical

One Categorical One Quantitative

Two Quantitative

Graphical Plot

Clustered bar chart crosstab

Comparative boxplot

Scatterplot

Histogram:

- Symmetric (even)
 - Skewness ~ 0
- Left skewed (trailing tail to the left)
 - Median $>$ mean
 - Negative skewness
- Right skewed (trailing tail to the right)
 - Median $<$ mean
 - Positive skewness
- Bimodal (two peaks)

Numerical Summary

Location a value most of the sample is centred around (mean, median, mode ...)

Variability how spread the values are around the centre

Location Measurements

Sample mean $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

Sample median $\tilde{x} = \begin{cases} \frac{1}{2}(x_{\frac{n}{2}} + x_{\frac{n}{2}+1}) & \text{even} \\ x_{\frac{n+1}{2}} & \text{odd} \end{cases}$ is the middlemost value of the sample

Quartiles The value such that some percent of observations are less or than equal to the value.
(0% = min, 50% = median, 100% = max)

Five number summary 0%, 25%, 50%, 75%, 100% quartiles.

Variability Measurements

Sample variance $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$

Sample Standard Deviation $s = \sqrt{s^2}$

IQR interquartile range = $q_3 - q_1$

an outlier is an observation farther than $1.5 \times \text{iqr}$ from the closest quartile.

Random Variables

Random Variable A random variable is a real valued function defined over the sample space:

$$\begin{aligned} X : S &\rightarrow \mathbb{R} \\ \omega &\rightarrow X(\omega) \end{aligned}$$

Usually, random variables are denoted by an uppercase letter.

Domain of Variation The domain of variation of X , S_X , is the set of all possible values that X can take.

X the name of the random variable

$X(\omega)$ the numerical value taken by the random variable at some sample point ω

x a generic numerical value

Cumulative Distribution Function The cdf (cumulative distribution function) of a random variable X is defined for any real number x by

$$F(x) = \mathbb{P}(X \leq x)$$

All probability questions about X can be answered in terms of its distribution.

Notation: $X \sim F$ reads X follows the distribution F .

- For any $a \leq b$, $\mathbb{P}(a < X \leq b) = F(b) - F(a)$

Discrete Random Variables

Discrete Random Variable A random variable is **discrete** if it can only assume a finite (or at most countably infinite) number of values.

Probability Mass Function The pmf of a discrete random variable X is defined for any real number x by

$$p(x) = \mathbb{P}(X = x)$$

The **Bernoulli random variable** is the simplest possible discrete random variable. $S_X = \{0, 1\}$ where $\mathbb{P}(0) = \pi - 1$ and $\mathbb{P}(1) = \pi$.

Continuous Random Variable

Continuous Random Variable A random variable X is **continuous** if there exists a nonnegative function $f(x)$ defined for all real $x \in \mathbb{R}$ such that for any set B of real numbers,

$$\mathbb{P}(X \in B) = \int_B f(x) \, dx$$

$f(x)$ is the **probability density function** (pdf).

Pdf Properties

- $F(x) = \mathbb{P}(X \leq x) = \int_{-\inf}^x f(y) \, dy$
- $f(x) \geq 0 \quad \forall x \in \mathbb{R}$
- $\mathbb{P}(a \leq X \leq b) = \int_a^b f(x) \, dx = F(b) - F(a)$
- $\int_{-\inf}^{+\inf} f(x) \, dx = 1$

Expectation and variance

Expectation The **expectation** or the **mean** of a random variable X , denoted $\mathbb{E}(X)$ or μ is:

Discrete

Continuous

$$\mu = \mathbb{E}(X) = \sum_{x \in S_X} xp(x)$$

$$\mu = \mathbb{E}(X) = \int_{S_X} xf(x) \, dx$$

The **expectation** has the same unit as X .

$\mathbb{E}(X)$ is a weighted average of all the possible values of X , with each value being weighted by the probability that X assumes it.

$\mathbb{E}(X)$ has the same units as X .

Expectation of a function of a random variable

Discrete

$$\mathbb{E}(g(X)) = \sum_{x \in S_X} g(x)p(x)$$

Continuous

$$\mathbb{E}(g(X)) = \int_{S_X} g(x)f(x) \, dx$$

Variance The **variance** of a random variable X , usually denoted by $\mathbb{V}\text{ar}(X)$ or σ^2 is defined by

$$\mathbb{V}\text{ar}(X) = \mathbb{E}((X - \mu)^2)$$

Discrete

$$\sigma^2 = \mathbb{V}\text{ar}(X) = \sum_{x \in S_X} (x - \mu)^2 p(x)$$

Continuous

$$\sigma^2 = \mathbb{V}\text{ar}(X) = \int_{S_X} (x - \mu)^2 f(x) \, dx$$

variance has the unit of X^2

Alternative formula:

$$\sigma^2 = \mathbb{V}\text{ar}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \mathbb{E}(X^2) - \mu^2$$

Standard Deviation The **standard deviation** of X , $\sigma = \sqrt{\sigma^2} = \sqrt{\mathbb{V}\text{ar}(X)}$

Standardisation

Suppose you have a random variable X with mean μ and variance σ^2 .

The associated **standardised** random variable, often denoted Z is given by

$$Z = \frac{X - \mu}{\sigma}$$

Thus:

$$\mathbb{E}(Z) = \frac{1}{\sigma} \mathbb{E}(X) - \frac{\mu}{\sigma} = \frac{\mu}{\sigma} - \frac{\mu}{\sigma} = 0$$

$$\mathbb{V}\text{ar}(Z) = \frac{1}{\sigma^2} \mathbb{V}\text{ar}(X) = \frac{\sigma^2}{\sigma^2} = 1$$

A standardised random variable always has **mean 0** and **variance 1**.

Jointly Distributed Random Variables

Joint Cumulative Distribution Function The **joint cumulative distribution function** of X and Y is given by

$$F_{XY}(x, y) = \mathbb{P}(X \leq x, Y \leq y) \quad \forall (x, y) \in \mathbb{R} \times \mathbb{R}$$

Joint Probability Mass Function If X and Y are both discrete, the **joint probability mass function** is defined by

$$p_{XY}(x, y) = \mathbb{P}(X = x, Y = y)$$

The **marginal pmf** of X and Y can be obtained by

$$p_{X(x)} = \sum_{y \in S_Y} p_{XY}(x, y)$$
$$p_{Y(y)} = \sum_{x \in S_X} p_{XY}(x, y)$$

Jointly Continuous X and Y are said to be **jointly continuous** if there exists a function $f_{XY}(x, y) : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}^+$ such that for any sets A and B of real number

$$\mathbb{P}(X \in A, Y \in B) = \int_A \int_B f_{XY}(x, y) \, dy \, dx$$

$f_{XY}(x, y)$ is the **joint probability density function** of X and Y .

The **marginal densities** follow from

$$\int_A f_{X(x)} \, dx = \mathbb{P}(X \in A) = \mathbb{P}(X \in A, Y \in S_Y) = \int_A \int_{S_Y} f_{XY}(x, y) \, dy \, dx$$

Thus,

$$f_{X(x)} = \int_{S_Y} f_{XY}(x, y) \, dy$$
$$f_{Y(y)} = \int_{S_X} f_{XY}(x, y) \, dx$$

The **expectation of a function of two random variables** $g : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is

$$\begin{aligned} \mathbb{E}(g(X, Y)) &= \sum_{x \in S_X} \sum_{y \in S_Y} g(x, y) p_{XY}(x, y) \\ &= \int_{S_X} \int_{S_Y} g(x, y) f_{XY}(x, y) \, dy \, dx \end{aligned}$$

Independent Random Variables

Independent Random Variables The random variables X and Y are said to be **independent** is, for all $(x, y) \in \mathbb{R} \times \mathbb{R}$,

$$\mathbb{P}(X \leq x, Y \leq y) = \mathbb{P}(X \leq x) \times \mathbb{P}(Y \leq y)$$

If X and Y are **independent**, then for any functions h and g

$$\mathbb{E}(h(X)g(Y)) = \mathbb{E}(h(X)) \times \mathbb{E}(g(Y))$$

Covariance and Correlation

Covariance of two random variables The **covariance** of two random variables X and Y is defined by

$$\text{Cov}(X, Y) = \mathbb{E}((X - \mathbb{E}(X))(Y - \mathbb{E}(Y)))$$

Properties

- $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
- $\text{Cov}(X, X) = \text{Var}(X)$
- $\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$
- $\text{Cov}(aX + b, cY + d) = ac\text{Cov}(X, Y)$
- $\text{Cov}(X_1 + X_2, Y_1 + Y_2) = \text{Cov}(X_1, Y_1) + \text{Cov}(X_1, Y_2) + \text{Cov}(X_2, Y_1) + \text{Cov}(X_2, Y_2)$

Interpretation

- If $\text{Cov}(X, Y) > 0$, X and Y tend to increase or decrease together.
- If X and Y are independent $\Rightarrow \text{Cov}(X, Y) = 0$, however $\text{Cov}(X, Y) = 0 \not\Rightarrow X$ and Y independent.
- $\text{Cov}(X, Y) = 0 \Rightarrow$ no linear association between X and Y .
- If $\text{Cov}(X, Y) < 0$, X tends to increase as Y decrease and vice-versa.

Variance of sum of r.v

$$\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab\text{Cov}(X, Y)$$

Correlation

Correlation Coefficient Covariance is important as an indicator of relationship between r.v's.

However, it depends on units of X and Y .

The **correlation coefficient** is often used instead.

$$\rho = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

- It is dimensionless.
- Always between -1 and 1 .
- Positive (and negative) ρ means positive (and negative) linear relationship.
- Closer $|\rho|$ is to 1 , stronger the linear relationship.

Special Random Variables

Binomial Distribution

Binomial Distribution Assume

- The outcome of a random experiment can be classified as either a success or a failure.
- We observe a success with probability π .
- n independent repetitions of this experiment are performed.

Define X = number of success observed over the n repetitions. We say that X is a binomial random variable with parameters n and π .

$$X \sim \text{Bin}(n, \pi)$$

$S_X = \{0, 1, 2, \dots, n\}$ and the binomial probability mass function is given by

$$p(x) = \binom{n}{x} \pi^x (1 - \pi)^{n-x} \quad \text{for } x \in S_X$$

where $\binom{n}{x}$ is the number of different groups of x objects that can be chosen from a set of n objects.

Bernoulli Distribution The Bernoulli distribution is a binomial distribution with $n = 1$.

$$X \sim \text{Bern}(\pi)$$

$$p(x) = \begin{cases} 1 - \pi & \text{if } x = 0 \\ \pi & \text{if } x = 1 \\ 0 & \text{otherwise} \end{cases}$$

If $X_1 = \text{Bin}(n_1, \pi)$, $X_2 = \text{Bin}(n_2, \pi)$ and X_1 and X_2 are independent, then

$$X_1 + X_2 \sim \text{Bin}(n_1 + n_2, \pi).$$

Mean and Variance of Binomial Distribution If $X \sim \text{Bin}(n, \pi)$,

$$\mu = \mathbb{E}(X) = n\pi \text{ and } \sigma^2 = \text{Var}(X) = n\pi(1 - \pi)$$

Poisson Distribution

Poisson Distribution Assume you are interested in the number of occurrences of some random phenomenon in a fixed period of time.

Define X = number of occurrences. We say that X is a Poisson random variable with parameter λ .

$$X \sim \mathcal{P}(\lambda)$$

if

$$p(x) = e^{-\lambda} \frac{\lambda^x}{x!} \quad \text{for } x \in S_X = \mathbb{N}.$$

Poisson distribution models a Binomial distribution where $n \rightarrow \infty$.

Let $X \sim \text{Bin}(n, \pi)$,

$$\begin{aligned}\mathbb{P}(X = x) &= \frac{n!}{x!(n-x)!} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x} \\ &= \frac{n!}{n^x(n-x)!} \frac{\lambda^x}{x!} \left(1 - \frac{\lambda}{n}\right)^n \\ \text{As } n \rightarrow \infty &= \frac{n!}{n^x(n-x)!} \rightarrow 1 \text{ and } \left(1 - \frac{\lambda}{n}\right)^n \rightarrow e^{-\lambda}\end{aligned}$$

Thus,

$$\mathbb{P}(X = x) = e^{-\lambda} \frac{\lambda^x}{x!} \quad \text{for } x \in \mathbb{N}.$$

λ is the **intensity** of the phenomenon.

Mean and Variance of Poisson Distribution If $X \sim \mathcal{P}(\lambda)$,

$$\mathbb{E}(X) = \lambda \quad \mathbb{V}\text{ar}(X) = \lambda$$

Uniform Distribution

Uniform Distribution A random variable is said to be **uniformly distributed** over an interval $[\alpha, \beta]$, i.e

$$X \sim U_{[\alpha, \beta]}$$

if its pdf(probability density function) is given by

$$f(x) = \begin{cases} \frac{1}{\beta - \alpha} & \text{if } x \in [\alpha, \beta] \\ 0 & \text{otherwise} \end{cases} \quad (\rightarrow S_X = [\alpha, \beta])$$

By integration

$$F(x) = \begin{cases} 0 & \text{if } x < \alpha \\ \frac{x - \alpha}{\beta - \alpha} & \text{if } \alpha \leq x \leq \beta \\ 1 & \text{if } x > \beta \end{cases}$$

Mean and Variance of Uniform Distribution

$$\mathbb{E}(X) = \frac{\alpha + \beta}{2} \quad \mathbb{V}\text{ar}(X) = \frac{(\beta - \alpha)^2}{12}$$

Exponential Distribution

A poisson distributed random variable counts the number of occurrences of a given phenomenon over a unit period of time.

If $N \sim \mathcal{P}(\lambda)$ denotes the number of occurrences over a unit period of time, then the number of occurrences of the phenomenon by a time x is $N_x \sim \mathcal{P}(\lambda x)$.

Denote X the amount of time before the first occurrence.

This time will exceed $x (x \geq 0)$ if and only if there have been no occurrences of the phenomenon by time x , that is, $N_x = 0$.

As $N_X \sim \mathcal{P}(\lambda x)$, it follows $\mathbb{P}(X > x) = \mathbb{P}(N_x = 0) = e^{-\lambda x} \frac{(\lambda x)^0}{0!} = e^{-\lambda x}$

X is called the Exponential distribution.

Exponential Distribution A random variable is said to be an **Exponential random variable** with parameter $\mu > 0$, i.e.

$$X \sim \text{Exp}(\mu),$$

if its probability distribution function is given by

$$f(x) = \begin{cases} \frac{1}{\mu} e^{-\frac{x}{\mu}} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases} \quad (\rightarrow S_X = \mathbb{R}^+).$$

By integration:

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 - e^{-\frac{x}{\mu}} & \text{if } x \geq 0 \end{cases}.$$

It is useful for representing random amounts of time, like the amount of time required to complete a task, waiting time at a counter, the amount of time until you receive a phone call, etc.

Note: the parameter $\mu = \frac{1}{\lambda}$ (λ = parameter of equivalent poisson distribution).

Mean and Variance of the Exponential Distribution If $X \sim \text{Exp}(\mu)$,

$$\mathbb{E}(X) = \mu \quad \text{Var}(X) = \mu^2.$$

Normal Distribution

Normal Distribution A random variable is **normally distributed** with parameters μ and $\sigma > 0$,

$$X \sim N(\mu, \sigma),$$

if its probability density function is given by

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (\rightarrow S_X = \mathbb{R})$$

No closed form exists for

$$F(x) = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^x e^{-\frac{(y-\mu)^2}{2\sigma^2}} dy$$

so numerical integration is required.

Mean and Variance of the Normal Distribution If $X \sim \mathcal{N}(\mu, \sigma)$,

$$\mathbb{E}(X) = \mu \quad \text{Var}(X) = \mu^2$$

Standard Normal Distribution is the normal distribution with $\mu = 0$ and $\sigma = 1$. Thus

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

Notation:

$$f(x) \doteq \varphi(x) \quad F(x) \doteq \Phi(x) \\ Z \sim N(0, 1)$$

Normal Distribution Standardisation

If $X \sim N(\mu, \sigma)$, then

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

68-95-99 rule:

$$\int_{-1}^1 \varphi(x) dx = \mathbb{P}(-1 < Z < 1) \approx 0.6827 \\ \int_{-2}^2 \varphi(x) dx = \mathbb{P}(-2 < Z < 2) \approx 0.9545 \\ \int_{-3}^3 \varphi(x) dx = \mathbb{P}(-3 < Z < 3) \approx 0.9973$$

68% of the data is within one standard deviation of the mean. 95% of the data is within two standard deviations of the mean. 99% of the data is within three standard deviations of the mean.

Normal Distribution Quantiles

Quantils Given $\mathbb{P}(Z > z_\alpha) = 1 - \alpha$ for $Z \sim N(0, 1)$, the value z_α is called the **quantile** of level α .

Normal Distribution Properties

If $X_1 \sim \mathcal{N}(\mu_1, \sigma_1)$, $X_2 \sim \mathcal{N}(\mu_2, \sigma_2)$ and X_1 is independent to X_2 , then for any $a, b \in \mathbb{R}$,

$$aX_1 + bX_2 \sim N\left(a\mu_1 + b\mu_2, \sqrt{a^2\sigma_1^2 + b^2\sigma_2^2}\right)$$